When exploration follows the wrong signal

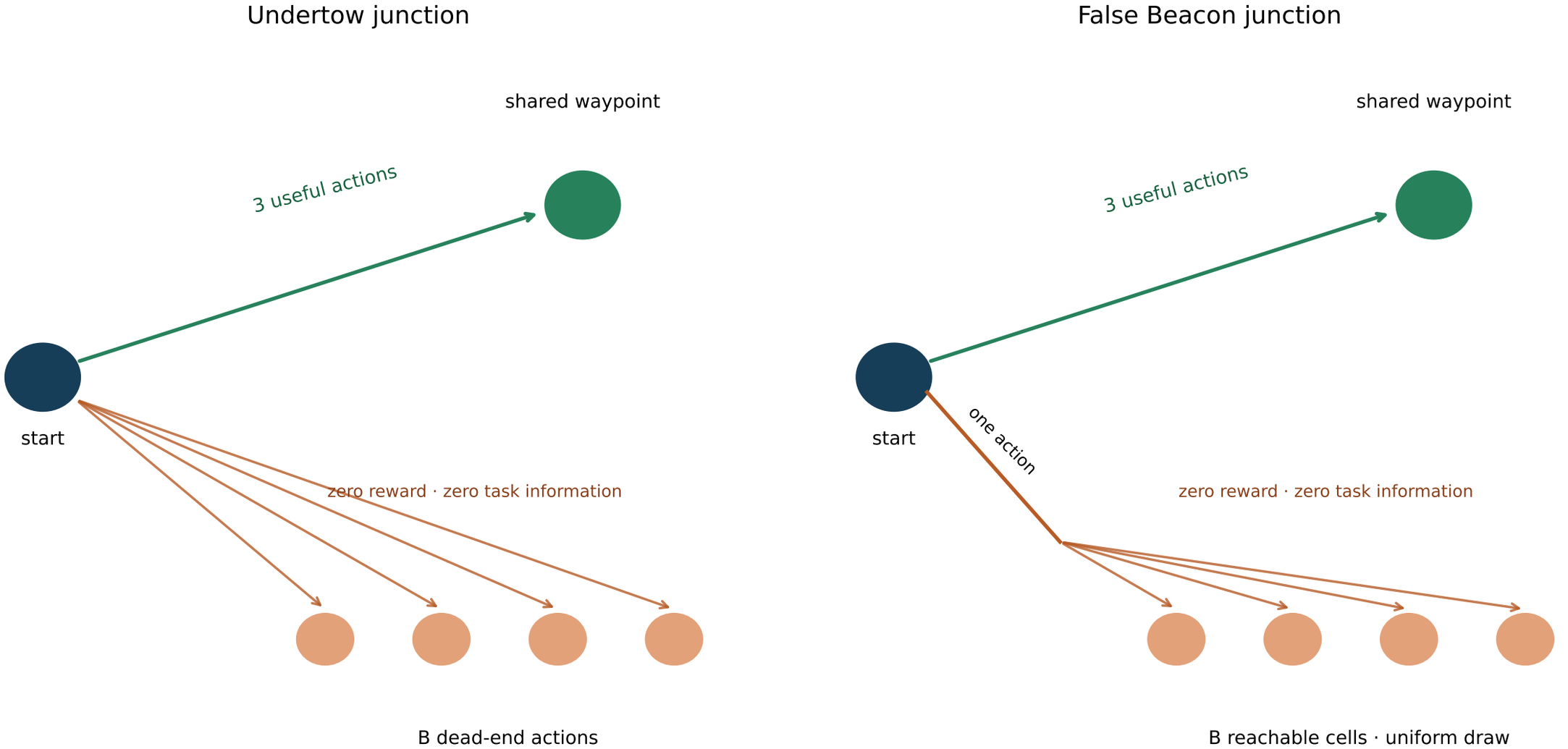
Question: does the same algorithm learn an easy problem, then deteriorate as irrelevant exploration opportunities or misleading cue strength increase? The figures follow STEERING's within-algorithm variable-hardness visual logic.

Physical burden: Undertow adds zero-reward actions; False Beacon widens a zero-reward chamber. IDAC can favor irrelevant occupancy; optimism pays to resolve a growing domain. At Undertow B=64, POMD stays uniform because its optimistic cost-Q values remain clipped at zero.

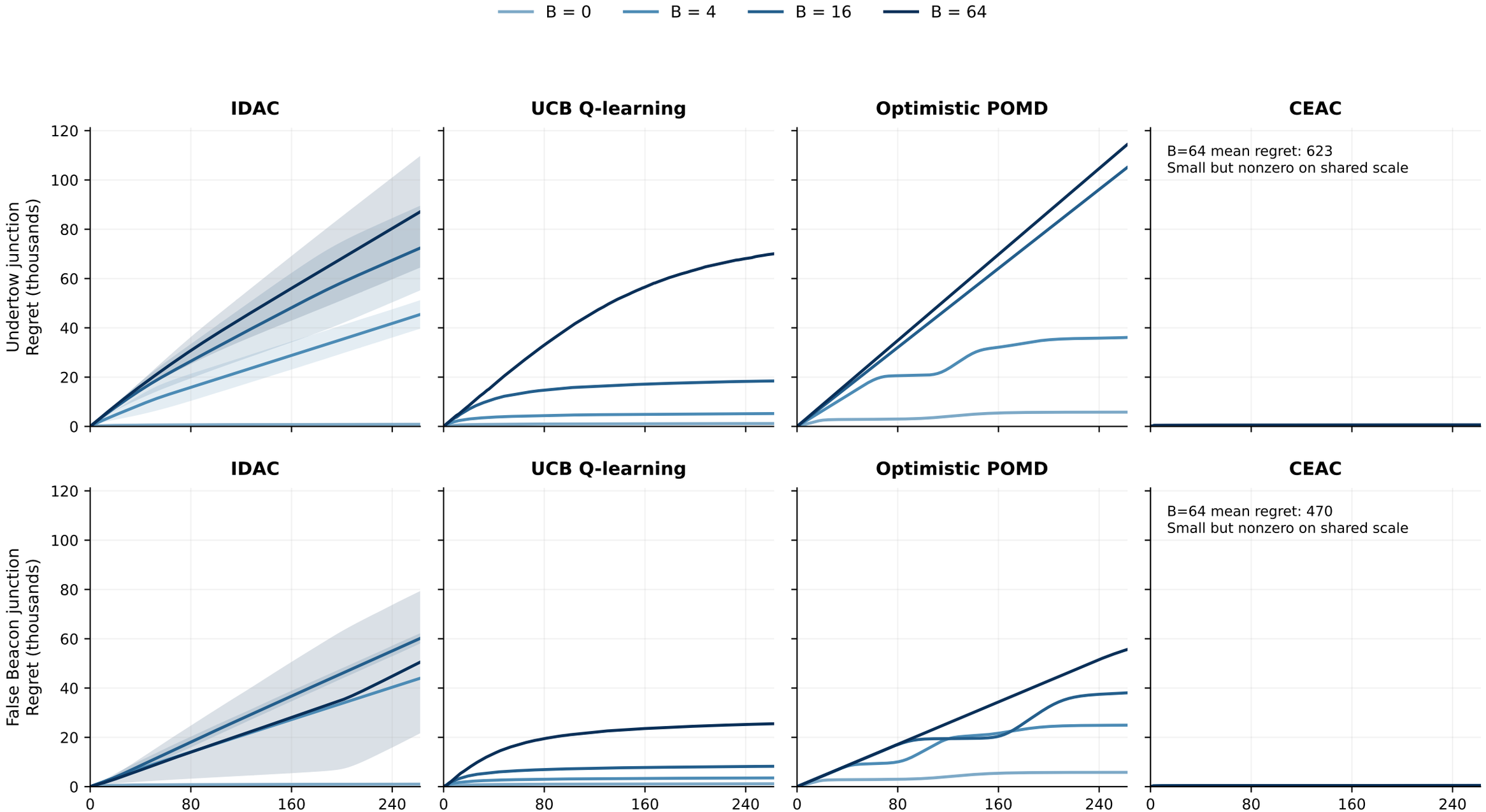
Prior misspecification: hold the Bayesian prior fixed while reversing the task–cue correlation. The true joint law leaves support, although both task labels remain possible. Correlated-prior IDS fails at full reversal; product-support IDS survives.

CEAC uses the requested reward critic, price-conditioned evidence critic and evidence-priced replay. In the original three-seed slate it avoids the large burden cost. Its task-0 actor-learning failures remain visible; correct evidence does not guarantee successful finite-budget policy optimization.

Scope: controlled two-step junction variants, not the historical full grids. Every dose is a separate stationary run. Primitive controls observe trajectories; CEAC/IDS additionally know the candidate likelihood laws. The oracle additionally knows the truth. Shared figure scales and all held-out seeds are retained.

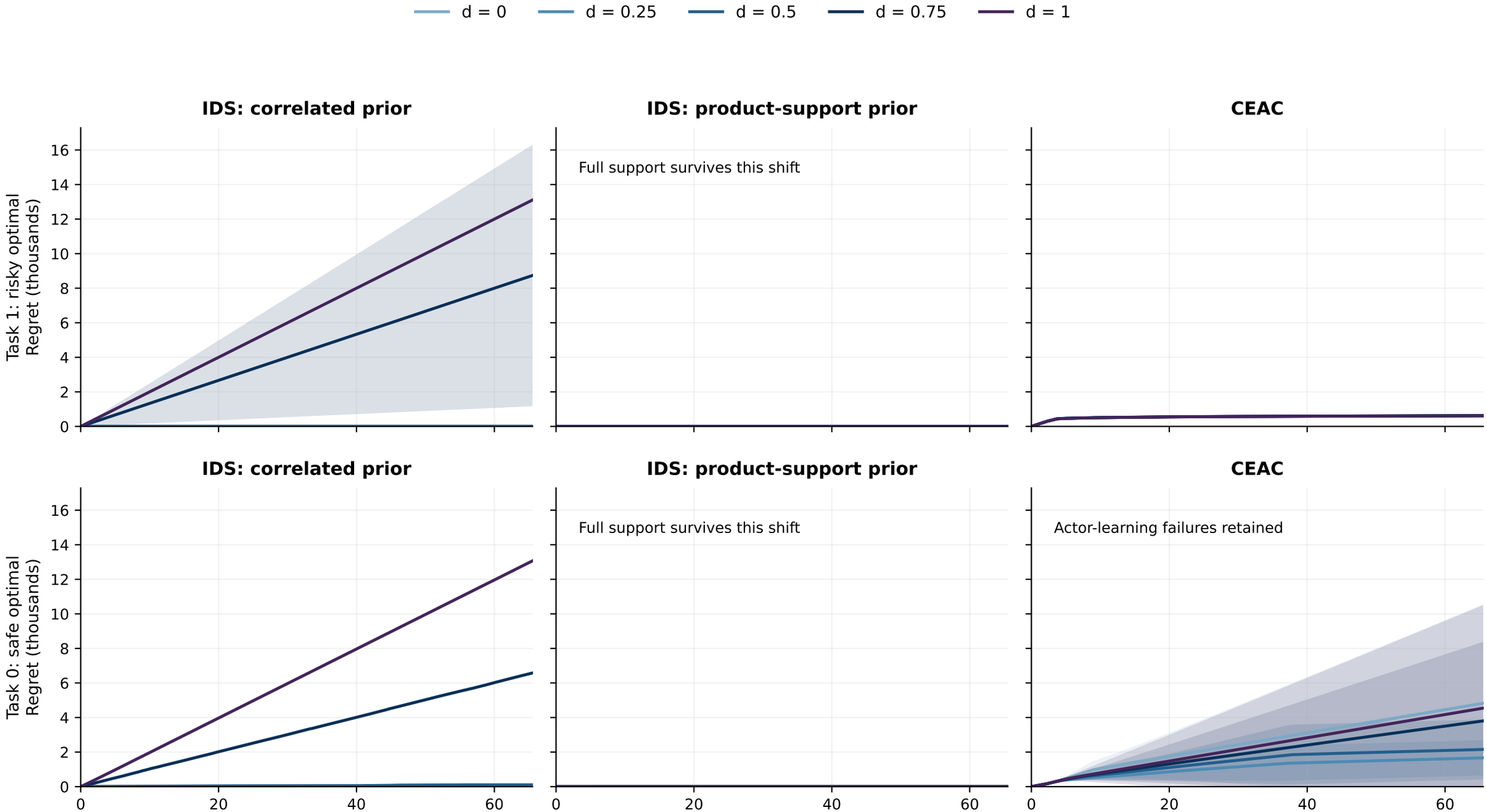


Each episode has two physical steps; useful rewards and task information remain fixed as B grows.



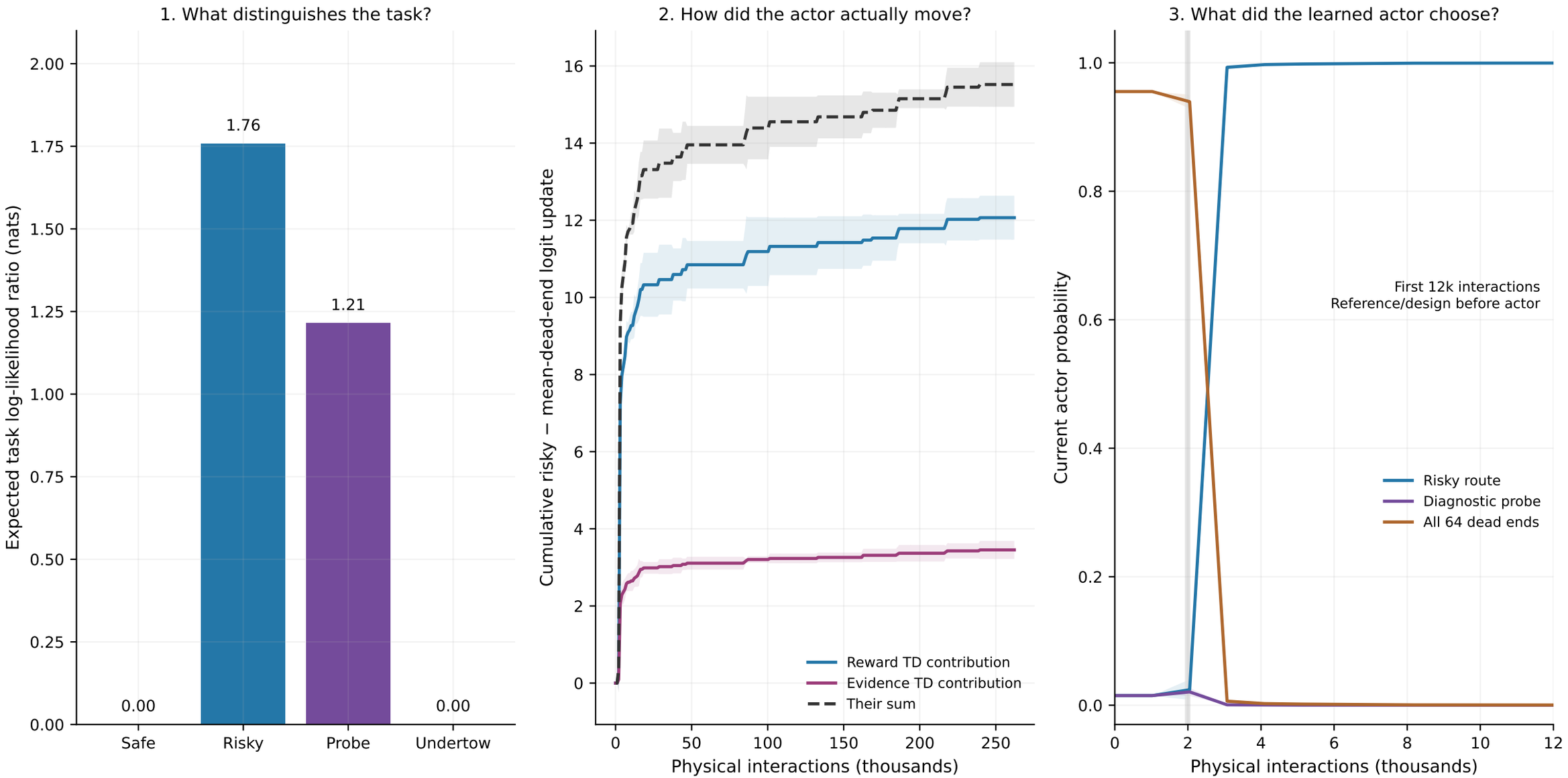
Same algorithm, increasing burden. Mean +/- sample SD over seeds 0, 1, 2; shared linear scales.
Full-support IDS and oracle also succeed (mean regret <= 0.694); all-method panels remain in the appendix.

Physical training interactions (thousands)

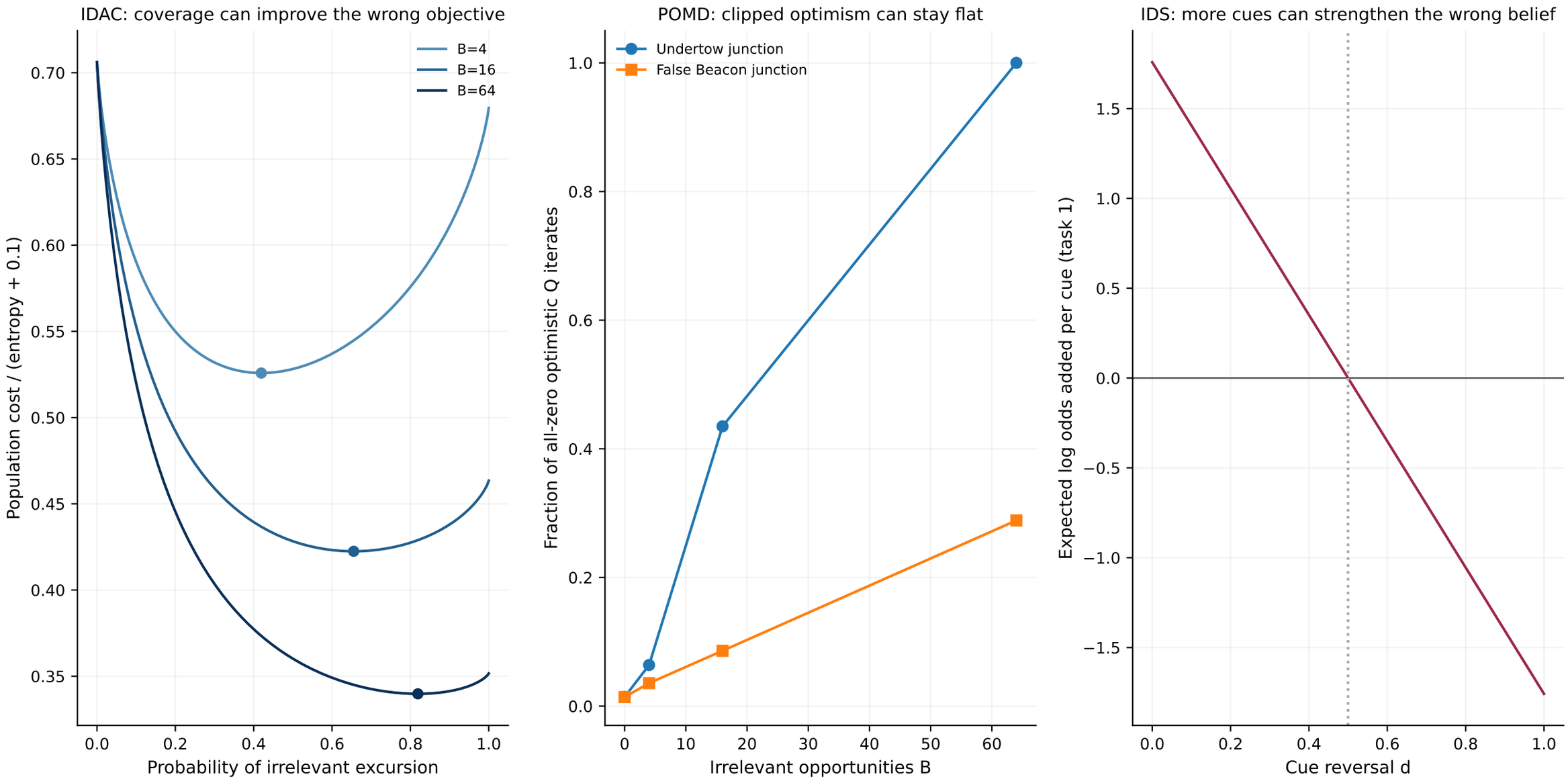


Same algorithm, increasing cue reversal. Mean +/- sample SD; shared scales.
Correlated-prior IDS fails at strong reversal; full-support IDS survives. CEAC task-0 failures are retained.

Physical training interactions (thousands)



Hard Undertow, B=64, all three seeds. Exact kernel KL and recorded actor-update decomposition.
No counterfactual performance claim. The right panel zooms to the first 12,000 interactions.



Supporting mechanism checks. Left: population objective, not a convergence claim (dots are minimizers).
Middle: actual optimistic-Q saturation. Right: analytic cue-only log evidence under the frozen prior.

What the evidence critic contributed — and what it did not

Measured contribution: at Undertow B=64, the cumulative risky-minus-mean-dead-end logit increment decomposes into 12.07 from reward TD and 3.45 from evidence TD, averaged over seeds. Their sum is the actual actor increment. At False Beacon B=64 the analogous evidence increment is 2.84. Here reward and task discrimination favor the same risky route. Positive evidence contribution reinforces that route; it is not a measured counterfactual improvement in final return.

Timing matters: the first certified replay design precedes or coincides with the first actor episode (episode 972 in these hard runs, zero-based). Thus initial reference identification is achieved by statistical replay before the learned evidence critic updates. The performance curves support the combined CEAC mechanism; they do not establish that the critic caused that initial identification.

Replay really uses evidence prices: for every logged design we verify `priced score = empirical return + beta × empirical likelihood-ratio evidence`. The audit reconstructs all 3,795,360 actor/two-critic updates across 54 CEAC runs, checks every recorded actor-policy identity, and checks that reward/evidence increments sum to the actual update (maximum per-step residual 9.2e-16). This uses saved observations, with no new training or alternative policy trajectory.

Task-0 failure: saved updates reproduce the correct reference and finite critic values. In the failed d=0/seed-0 run, cumulative risky-minus-safe logit contributions are -7.28 from reward TD and +15.22 from evidence TD. A substantial actor optimization residual remains. All seeds and failures are retained; no held-out tuning or actor-only ablation was performed.

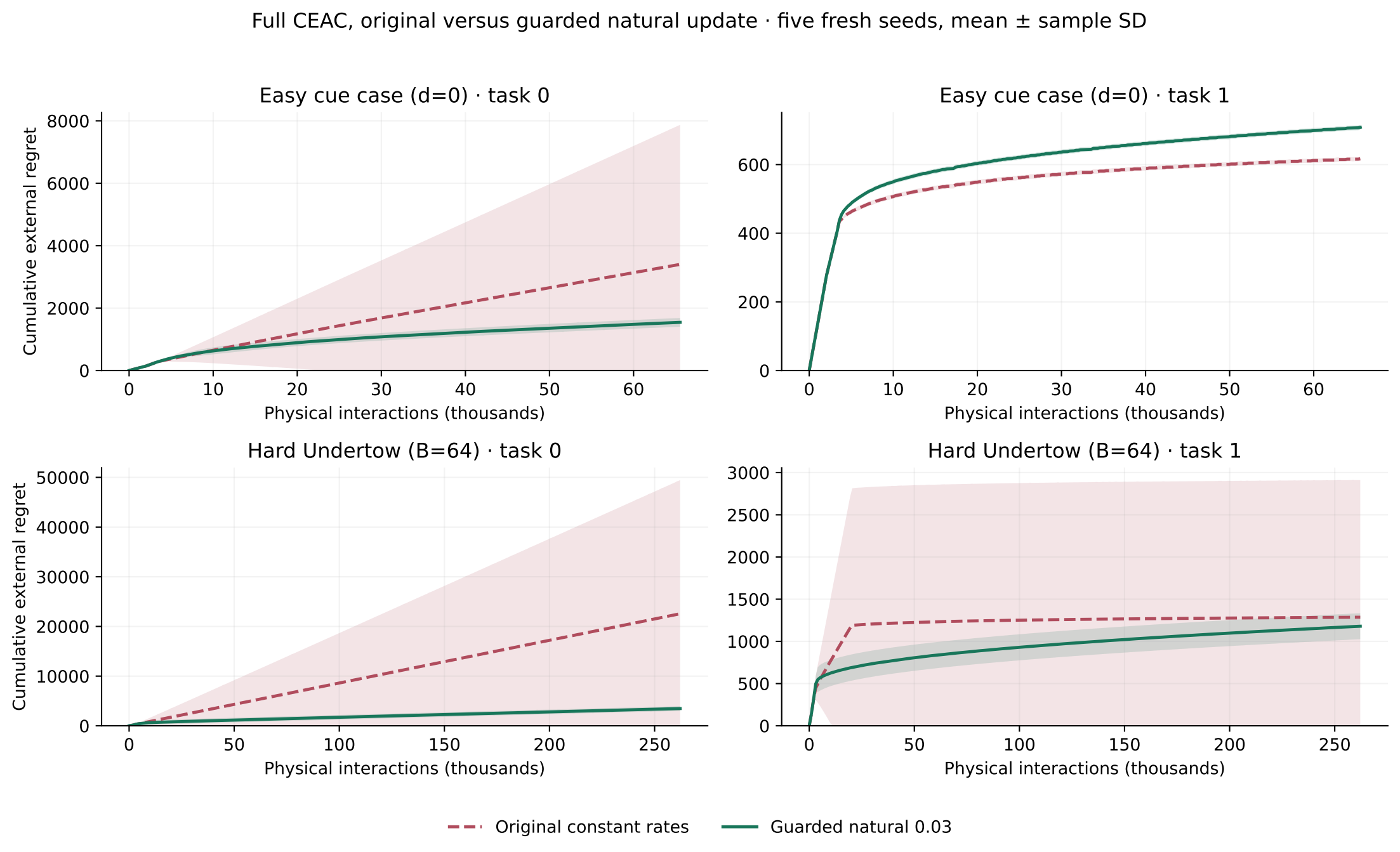
Read report.md / the executable notebook for full numerical tables, replay-score checks, the complete assumptions and links to all-dose/all-method evidence. These diagnostics decompose actual updates; they do not estimate final performance under removal of the critic.

Actor stability: speed and final accuracy differ

All settings retain both critics and evidence-priced replay. Original CEAC won the frozen endpoint-first ranking. Guarded natural 0.03 also passed calibration with about 49% lower regret. A secondary comparison of every non-reference passer was registered before inspecting fresh-seed outcomes; the primary selection did not change.

Family	Task	Original regret	Natural regret	Paired difference +/- SD
False Beacon	0	4,438	2,979	-1,460 +/- 3,551
False Beacon	1	473	686	213 +/- 10
Undertow	0	8,395	3,162	-5,233 +/- 6,449
Undertow	1	722	895	174 +/- 359
Cue reversal	0	2,224	1,523	-702 +/- 2,060
Cue reversal	1	618	712	94 +/- 1

Differences are natural minus original. Doses are averaged within each of five fresh seeds before computing SD. Cue runs use 32,768 episodes; burden runs use 131,072. The floor adds a real cost. Schedules, critic scaling and geometry change jointly, so their individual causal contributions are not isolated. External comparators used different seeds and retain their original results. All doses and endpoints appear in the stability assessment.


Full CEAC, original versus guarded natural update · five fresh seeds, mean ± sample SD
Easy cue case (d=0) · task 0
Easy cue case (d=0) · task 1
Hard Undertow (B=64) · task 0
Hard Undertow (B=64) · task 1
Cumulative external regret
Physical interactions (thousands)
Original constant rates
Guarded natural 0.03